

Scaling the pan/tilt down to a 28BYJ-48

Six modifications to ASSEMBLY_pan_tilt.stp, and why each one is forced. Target: FDM-printed, ~15 g sensor/pointer payload, pan travel unconstrained.

Where this starts

The source file is a 25-occurrence CATIA AP214 export with two verified 3:1 spur stages (module 1.125, 20T/60T), a 306 x 298 x 304 mm envelope, two unnamed placeholder motors, and four genuine fit clashes. It is a reference model, not a released design. What follows keeps its architecture — two perpendicular axes, 3:1 reduction on each, a yoke carrying the payload — and rebuilds everything the 28BYJ-48 touches.

The result is 110 x 126 x 85 mm: 2.8x smaller linearly, 24x smaller by bounding box volume.

This is not a scale factor. Three things in the original do not scale: the motor's shaft is offset from its can, the gear module has a printability floor, and bearings only exist in discrete sizes. Each forces a local redesign rather than a multiply.

1. What the motor dictates, before any geometry

Every downstream choice falls out of four numbers on the 28BYJ-48 datasheet.

Parameter	Value	What it forces
In-traction / self-positioning torque	> 34.3 mN.m	The 3:1 stage stays. It is the only source of torque.
Internal reduction	1/64 (nominal)	Resolution is already solved: 0.088 deg/step at the motor output.
Practical output speed at 5 V	~15 RPM	5 RPM at the axis = 30 deg/s. Fine for pointing, useless for tracking.
Output shaft	dia 5.0, double-D, offset 8 mm	Both motor mounts get redrawn from scratch. See section 4.

The 3:1 stage earns its place — but not for the reason you would guess

It is not there for resolution. 5.625/64 already gives 0.088 deg per step at the motor output; the 3:1 only sharpens that to 0.029 deg, which nothing in this payload class needs. It stays for two other reasons:

- **Torque.** 34.3 mN.m becomes 103 mN.m at the axis. Without it there is no margin.
- **Load path.** The 28BYJ-48's internal gear train is plastic. A payload hung directly off that output shaft puts radial and shock load straight into those gears, and they strip. The external stage takes the load instead; the motor only ever sees pinion tooth force.

Figure 7 — Tilt axis torque budget (pan sees no gravity torque at all)

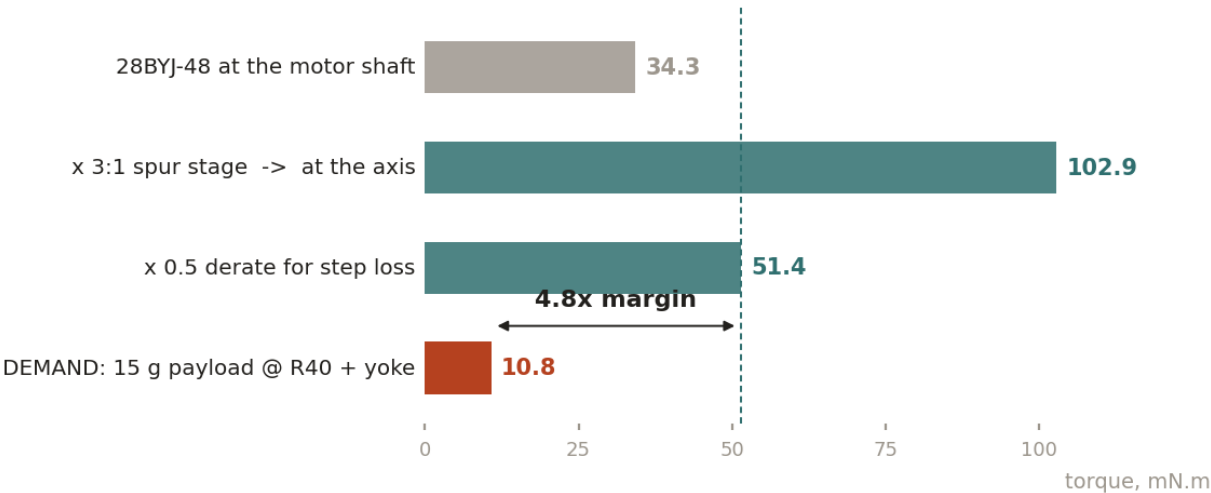


Figure 7 — 4.8x margin on the tilt axis. Pan sees no gravity torque at all: gravity produces zero moment about a vertical axis, so the pan motor fights only bearing friction and inertia.

The holding-torque question answers itself. 64:1 internal x 3:1 external = 192:1, plus >34.3 mN.m of self-positioning torque. At 15 g of payload the head holds position unpowered. No brake, no counterweight, no worm drive.

2. Modification one — collapse the third axis into two

The analysis of the source file found the Cam Holder's pivot bores lying perpendicular to, but not coaxial with, the gear-driven tilt axis, and could not resolve whether that was a deliberate third degree of freedom or a placement error. That ambiguity does not survive into a design you intend to build.

Decision: the yoke is the tilt member. It is keyed directly to the tilt shaft with a D-flat and an M3 grub screw. Two degrees of freedom, both driven, no orphan pivot. This also merges 'Cam Holder' and 'Baring 2 shaft' into one printed part.

3. Modification two — keep 3:1, drop the module

The ratio and the tooth counts are the two things worth keeping verbatim: 20T/60T at 20 deg pressure angle is clean, undercut-free, and already proven in the source geometry. Only the module changes.

Figure 2 — Gear stage re-scale: 3:1 ratio and 20T/60T are kept, module drops

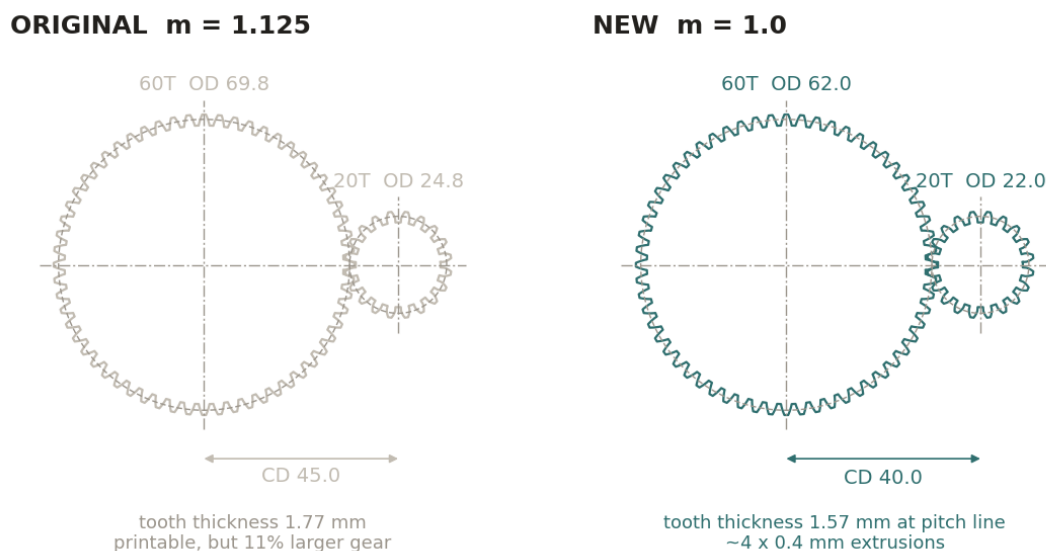


Figure 2 — Module 1.125 to 1.0. Centre distance 45.0 to 40.0 mm; the 60T gear goes from 69.75 to 62.0 mm OD.

Why 1.0 and not 0.8

Module 0.8 would give CD 32 mm and a 49.6 mm gear — another 8 mm of radius, which is tempting. But tooth thickness at the pitch line is $\pi m/2$: 1.26 mm at $m=0.8$ versus 1.57 mm at $m=1.0$. With a 0.4 mm nozzle that is roughly three extrusions per tooth against four. Three is where flank geometry starts depending on your slicer's opinion rather than yours, and backlash stops being predictable. At $m=1.0$ the tangential tooth load is about 3.4 N — nowhere near a strength limit — so the module is set purely by print fidelity, not stress.

If your printer is well tuned and you want the extra 8 mm, $m=0.8$ is a legitimate swap. Nothing else in the design changes except the centre distance.

Face width goes 5 to 6 mm — free, and it buys tolerance to a slightly misaligned printed bore.

4. Modification three — redraw both motor mounts

This is the modification that makes 'just scale it' impossible. The placeholder in the source file is a concentric can with a shaft down its centreline. The real 28BYJ-48 is not that part.

Figure 3 — The 28BYJ-48 will not drop into a scaled placeholder pocket

A — 28BYJ-48 face

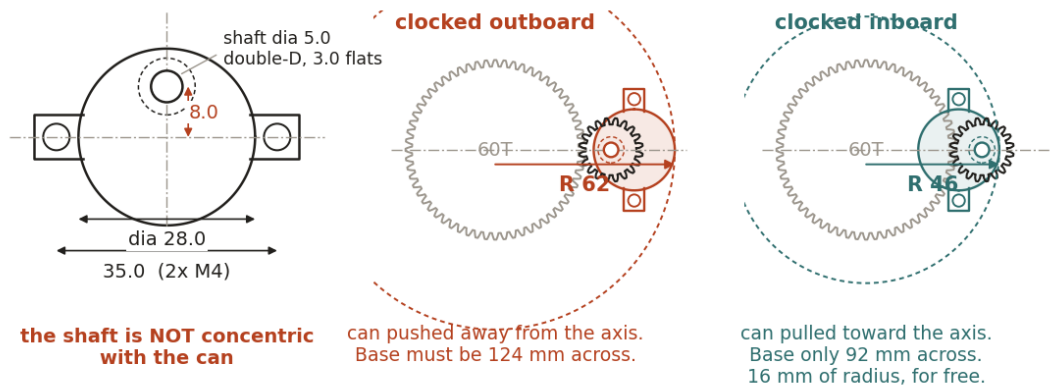


Figure 3 — The output shaft sits 8 mm off the can centreline, perpendicular to the line of the two M4 tabs. Clocking the motor so the can is pulled toward the gear axis, rather than pushed away, is worth 16 mm of radius and costs nothing.

Mount details

- **Bolt pattern:** 2x M4 on 35 mm centres. Print at 4.3 mm; M4 clearance in FDM wants the extra.
- **Can recess:** dia 29 (1 mm clearance on the 28 mm can), 19 mm deep or through.
- **Boss clearance:** dia 10 around the shaft boss.
- **Pinion bore:** dia 5.0 with the double-D flats at 3.0 mm. Model the flats — do not print a round bore and rely on a grub screw. The flats are what carry torque; a grub screw on a 5 mm plastic-geared shaft will round it off.
- **Clock both motors offset-inward**, as in Figure 3C.

Where each motor goes

Pan motor: can up, shaft down. Mounted to a bracket inside the base with the shaft pointing downward into the pan gear. If you mount it shaft-up the can has to live either below the base plate (which needs 19 mm of standoff) or inside a taller base. Inverting it keeps the base at 50 mm.

Tilt motor: 40 mm behind the tilt axis, not below it. The pinion axis can sit anywhere on a 40 mm circle around the tilt axis. Putting it directly below drives the whole head 50+ mm taller, because the can then has to clear the base as the platform pans. Putting it behind costs depth — which is the cheapest dimension here — and nothing else.

5. Modification four — re-select the bearings

The original's bearing sizes are meaningless at this scale, and one of them (the four-bolt flange, dia 25 bore) is the single part in the file that fails B-rep validity. All of it goes.

Original	New	Reasoning
Four-bolt flange, dia 25 bore (invalid topology)	2x 608ZZ, 8 x 22 x 7, on a dia 8 shaft	Two bearings 16 mm apart give moment stiffness a single flange bearing never had. 608 is the most available bearing on earth.
2x Wall Ball Bearing, dia 37.25 bore	2x 625ZZ, 5 x 16 x 5, on a dia 5 shaft	Tilt shaft carries ~0.15 N of payload weight. 5 mm is structurally ample and matches common rod stock.
2x 7804K114 (on the motor shafts)	deleted	The 28BYJ-48's shaft is supported inside the gearbox. A 6 mm pinion overhang at 3.4 N does not need a bearing.

Use ground steel rod for both shafts — 8 mm and 5 mm, cut to length. Printed shafts go oval and wear into the bearing bore within a few hundred cycles. This is the one place where FDM is simply the wrong process.

6. Modification five — fix the fits

The source file has four real clashes totalling about 1,417 mm³: two bearing-vs-shaft (538 and 514 mm³) and two bearing-vs-housing (330 mm³ each). They look like four problems. They are one problem, twice.

Figure 4 — Bearing seat, section through a 608ZZ (8 x 22 x 7)
WRONG — what the original file does **RIGHT — one race, one job**

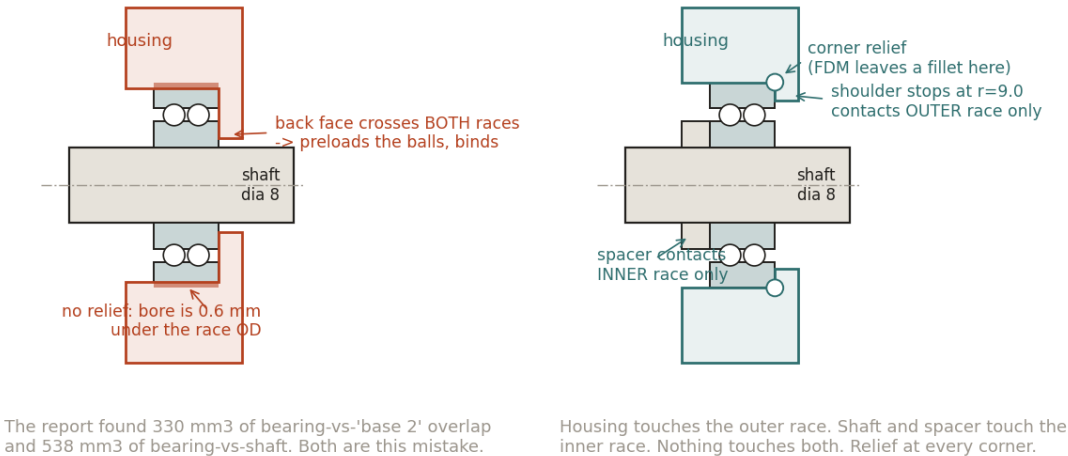


Figure 4 — One race, one job. The housing touches only the outer race; the shaft and its spacer touch only the inner race; nothing touches both. Every internal corner gets a relief.

What changes in practice

- **The 0.0 mm fit goes.** The original pan bearing is a 25.0 shaft in a 25.0 bore — line-to-line, no margin, and on the one part with invalid topology. In the new design the bearing is a bought part and defines its own fit.
- **Bearing pockets: print a coupon first.** FDM prints holes undersize by a process-dependent amount. Print a test block with 21.8 / 22.0 / 22.2 mm bores, pick the one that gives a firm press on a 608, and use that number everywhere. This is the single dimension worth calibrating before you commit to a full print.
- **Relief at every internal corner.** FDM leaves a fillet where a wall meets a floor, and that fillet will hold the bearing out of square. A 0.9 mm relief circle in the corner costs nothing and removes the problem.
- **Shoulder height < race thickness.** On a 608 the outer race is about 2.8 mm thick, so a housing shoulder reaching in to r=9.0 catches the outer race and clears the inner. Reach further and you preload the balls.

7. Modification six — delete two-thirds of the part count

17 unique part definitions become 11, and 25 occurrences become 19. Most of the deletions are things the source file only contained because it was an as-imported concept assembly.

#	Part	Qty	Make	Note
1	Base shell	1	print	Houses pan gear, both 608s, pan motor. ~94 mm across.
2	Pan motor bracket	1	print	Can-up mount. Merges the old 'Motor base'.
3	Pan platform + walls	1	print	Merges old 'Gear 2 Base'. Carries both 625s.
4	Yoke	1	print	Merges 'Cam Holder' + 'Baring 2 shaft'. D-bore, M3 grub.
5	Tilt motor bracket	1	print	Hangs 40 mm behind the tilt axis.
6	Spur gear 60T, m1.0, FW6	2	print	dia 8 D-bore (pan) / dia 5 D-bore (tilt).
7	Spur gear 20T, m1.0, FW6	2	print	dia 5 double-D bore, 3.0 flats.
8	28BYJ-48 + ULN2003	2	buy	Measure yours. See section 9.
9	608ZZ	2	buy	Pan shaft.
10	625ZZ	2	buy	Tilt shaft.
11	Steel rod dia 8 / dia 5	1 ea	cut	Pan and tilt shafts. Flat filed for the grubs.

Gone: the four-bolt flange bearing, both 7804K114s, both motor-body and motor-shaft placeholders, 'Motor base', 'Gear 2 Base', 'Baring 2 shaft', and all four 'Spacer' parts — which become printed bosses on the parts they used to sit between.

8. The result

Figure 1 — Envelope, front view, to scale

Original ASSEMBLY_pan_tilt.stp vs the 28BYJ-48 redesign

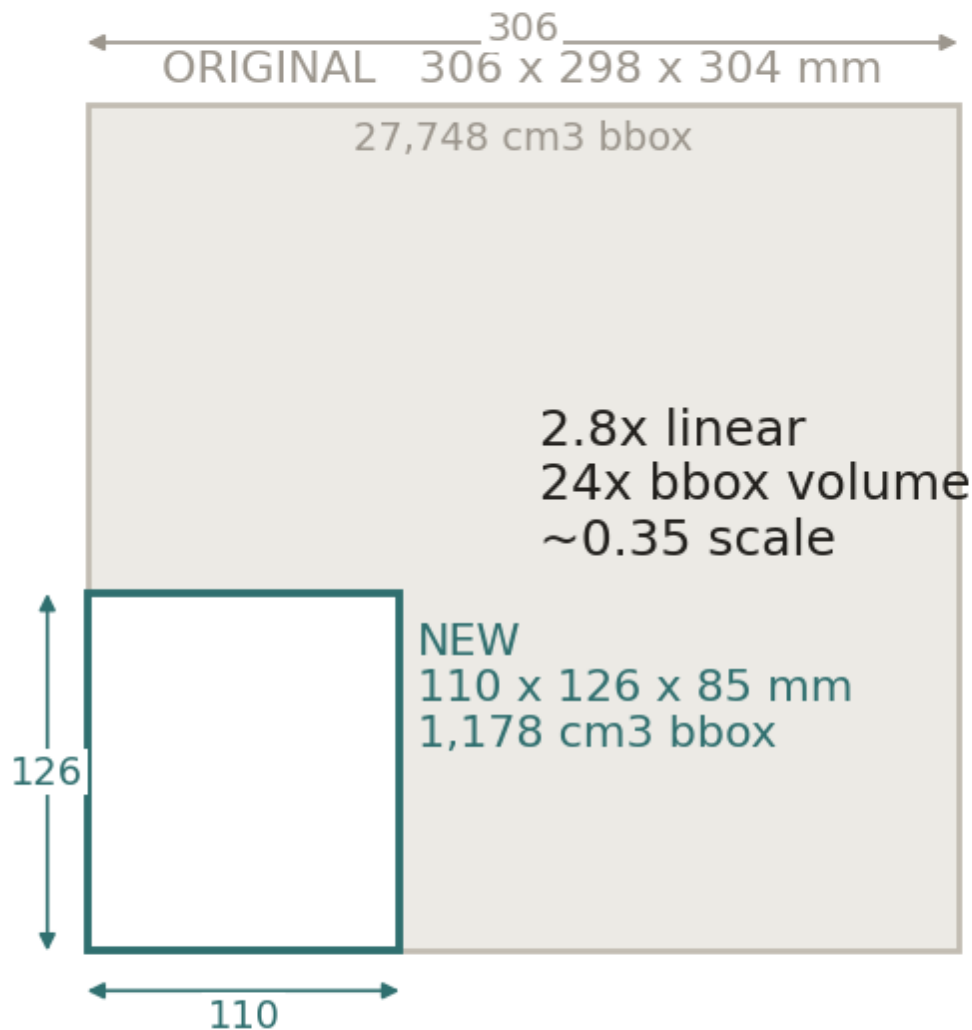


Figure 1 — Envelope, to scale.

	Original	New	Change
Envelope (mm)	306.5 x 298.1 x 303.7	110 x 126 x 85	2.8x linear
Bounding box volume	27,748 cm ³	1,178 cm ³	24x smaller
Unique parts / occurrences	17 / 25	11 / 19	–35% / –24%
Gear module	1.125	1.0	CD 45 → 40
Reduction (both axes)	3:1	3:1 (x 64:1 internal)	unchanged externally
Resolution at the axis	not determinable (no motor)	0.029 deg/step	—
Axis speed	not determinable	~5 RPM (30 deg/s)	—
Rotation axes	2 or 3 (ambiguous)	2 (resolved)	—
Interference volume	~1,417 mm ³ real clashes	0 by construction	—

General arrangement

Figure 5 — New general arrangement, front view

Layout schematic to scale, not a manufacturing drawing

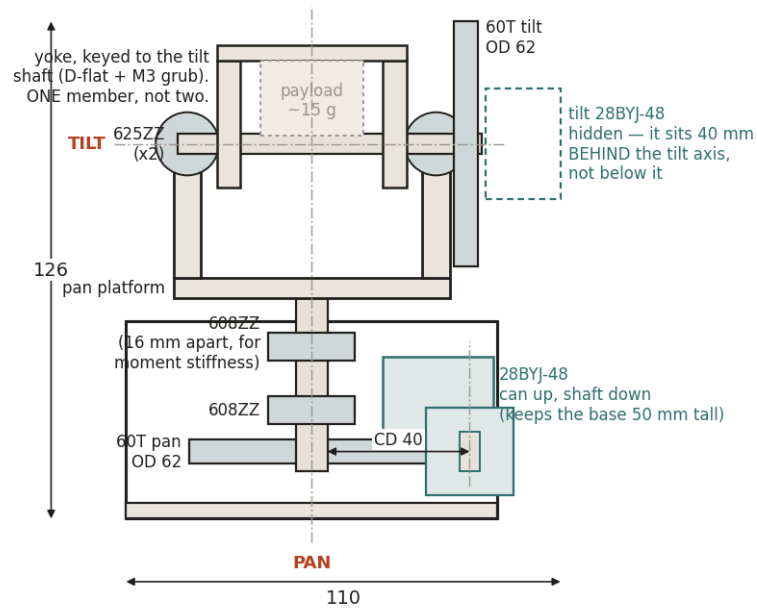


Figure 5 — Front view.

Figure 6 — Side view, looking along the tilt axis

The tilt motor tucks behind the gear — it costs depth, not height

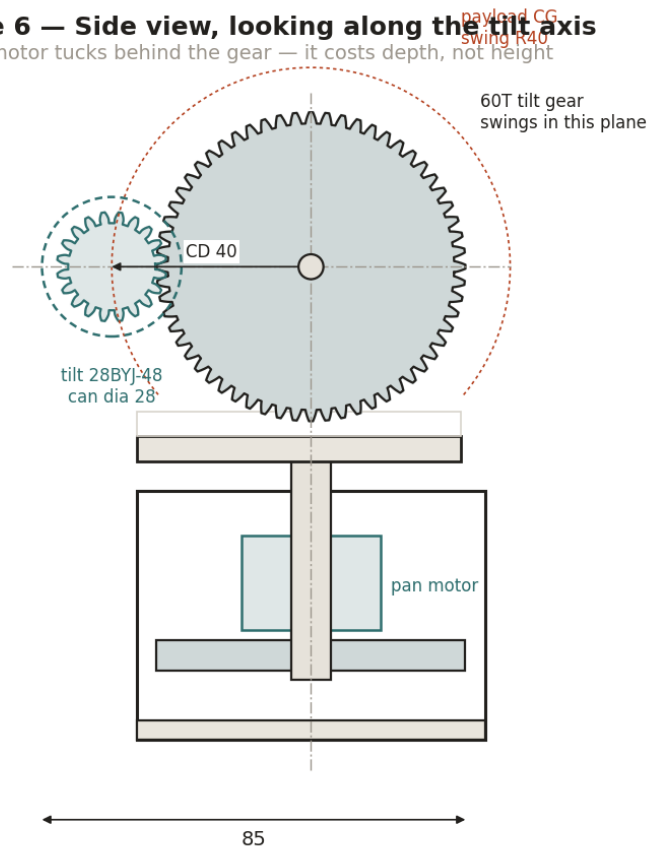


Figure 6 — Side view.

9. Print notes

- **Gears: print flat**, axis vertical, on the bed. The layer lines then run across the flank rather than along it. Printed on their side the teeth are unusable.
- **0.4 mm nozzle, 0.15 mm layers, 4 perimeters on gears**. At $m=1.0$ four perimeters is essentially a solid tooth, which is what you want.
- **PLA is fine**. Stiffer than PETG and the loads here are trivial. Use PETG only if it lives somewhere warm or outdoors.
- **Yoke: 4 perimeters, 30% infill**. It is the only part carrying a bending load, and even that is about 0.15 N.
- **Do not print the shafts**. See section 5.

10. What I would verify before cutting metal

Stated plainly, because several of these could invalidate a pocket dimension.

Biggest risk: measure your actual motor. The 8 mm shaft offset and 19 mm can height here come from vendor datasheets. The 28BYJ-48 is a widely cloned part and dimensions vary between suppliers. Calipers on the motor in your hand, before you commit the can recess and the 35 mm bolt pattern to a print.

- **The 1/64 ratio is nominal.** The common 28BYJ-48 is widely reported to be 63.68395:1, not 64:1 — the internal tooth counts do not divide evenly. That is a 0.5% error, about 1.8 deg accumulated over one full output revolution. Irrelevant for relative moves; fatal for absolute pointing by step-counting. Use the real ratio and add a home switch on each axis. Verify the ratio on your specific motor by counting steps against a full revolution.
- **Consider the 12 V variant.** Same frame, same mounting, roughly double the torque. If you can spare 12 V, margin goes from 4.8x to about 10x for free. The 5 V part is only the default because it is what ships with a ULN2003 board.
- **Backlash.** A printed 20T/60T pair at $m=1.0$ will show perhaps 0.3–0.5 deg of backlash at the output — an order of magnitude more than the 0.029 deg step. Your step resolution is a fiction unless you always approach a setpoint from the same direction. Design the control that way from the start.
- **Tilt gear to platform clearance is 3 mm** in this layout (gear bottom at $y=64$, platform top at $y=61$). That is the tightest clearance in the design. Check it in CAD before printing.
- **No FEA, no thermal.** At 3.4 N of tooth load I do not believe this needs either, but I have not run them. The 28BYJ-48 does get warm on continuous hold current — which, given the 192:1 self-holding, you should not need.
- **Pan travel is limited by the tilt motor's cable**, not by anything structural. You said this does not matter; if it starts to, a 4-wire slip ring at the pan axis is about 20 mm tall and the base has room for it.

What did not change: two perpendicular axes, 3:1 on each, 20T/60T at 20 deg PA, a yoke carrying the payload, and both stages sharing a ratio so pan and tilt have identical resolution. That architecture was the good part of the source file, and it survives intact at 0.35 scale.